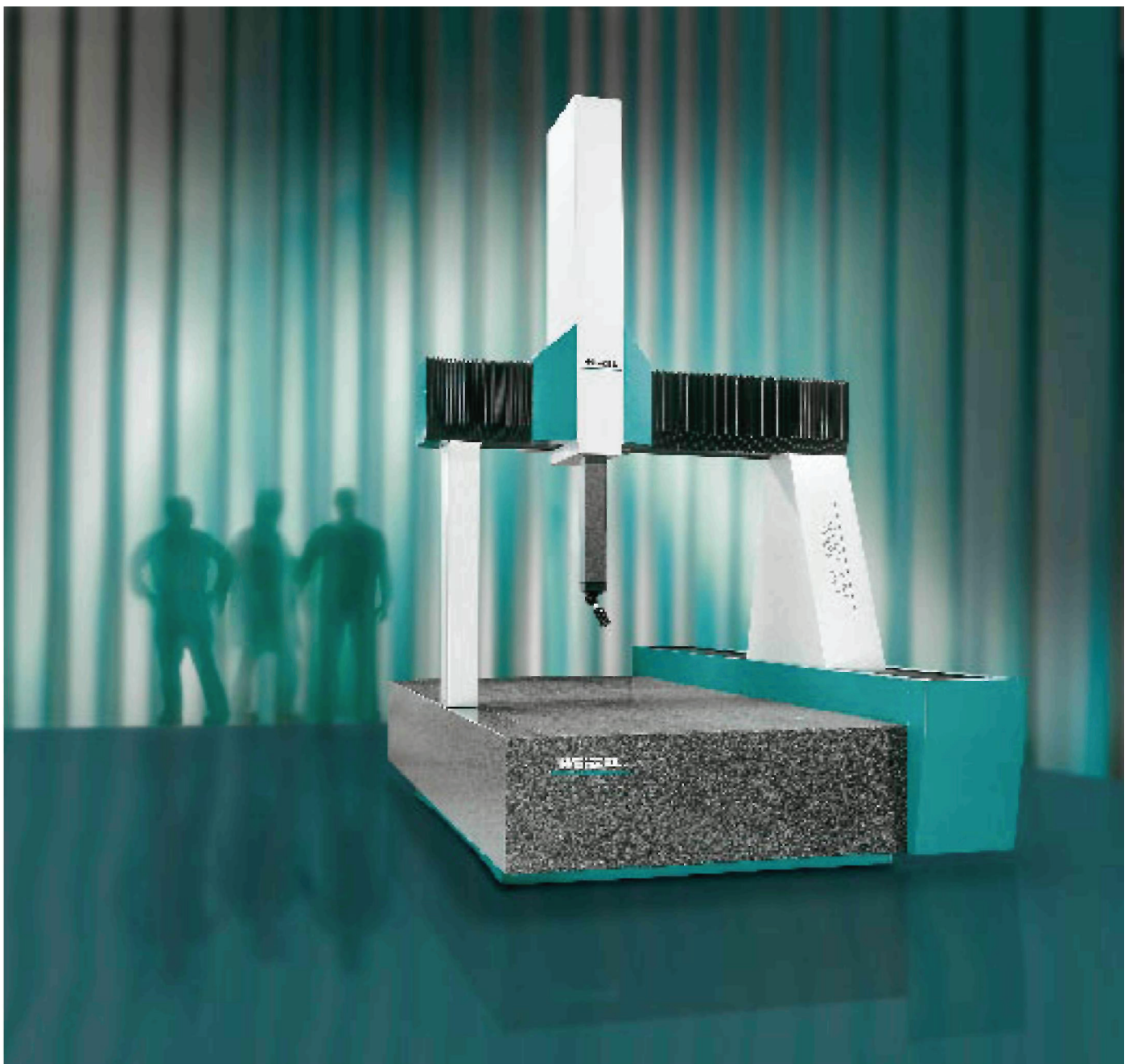


Co-ordinate Measuring Machines LH 1210, LH 1512 PREMIUM-SELECT Models

Technical Data



Technical Data LH 1210, LH 1512 PREMIUM-SELECT

Short description

- CNC-bridge design measuring machine with touch-trigger or scanning probe systems
- All granite guide-ways are accurately hand-lapped
- Compact design operator workstation, with integrated controller and computer
- CMM available in multiple sizes for the optimal selection of the required measurement volume

Application areas

- In production, incoming inspection and quality assurance
- Measurement of prismatic and free-form components
- Both series and individual measurements
- Palletised operation possible

Standard Features

- The Y-axis guide-way is machined directly in the base plate, providing optimal long-term stability
- Pre-stressed, encompassing air bearings in all axes
- Active pneumatic vibration damping
- Compact HT 400 control panel with central, logarithmic joystick, "mouse function" and context-sensitive function buttons. Selectable joystick's axis assignment
- The X- and Y- guideways feature bellows protections against contamination
- High-speed-dynamic servo drives with position monitoring, combined friction power transmission
- Three-axis contouring controller with intelligent "look-ahead" function for application-optimised trajectory
- Automatic on-line temperature compensation with sensors on machine axes and workpiece
- Two-stage speed selection and variable speed adjustment (override 0-100%) in all operation modes, resulting in sensitive movement via joystick or in CNC debugging

Probe systems

- Renishaw PH6, simple fixed probe head
- Renishaw PH10M/PH10MQ/PH10T, indexable probe heads featuring 720 repeatable positions in 7.5° steps
- SP25 scanning and single-point probe, ultra-precise and flexible for stylus lengths of up to 200 mm, optionally up to 400 mm. Probe module and stylus can be changed via optional tool changer. Use with PH10M or PH6M. For scanning and single-point probing
- Renishaw PH6M, like PH6, but with the possibility of using more complex probes (SP25, SP600)
- SP80 scanning probe head, ultra-precise for probe lengths up to 500 mm
For scanning and single-point probing. Stylus combinations can be changed via optional tool changer

Software

- User-friendly Windows software Metrosoft CM for measuring and evaluating geometry and free-form elements (option)
- Graphic user interface featuring extensive automatisms to support the User
- User dialog and reporting can be selected and switched on-line independently between 12 languages
- Graphically interactive on- and offline programming system "Grips" for measurement program creation based on CAD data.
- Numeric and graphic reporting of the measured results
- Work piece-oriented database, SQL-capable, with multi-user access, network-capabilities
- Integrated statistic functions, frequency distribution, trend diagram, machine-capability Cm and Cmk, SPC control charts, process capability Cp and Cpk. Interface to QS-Stat
- Shape- and location tolerances according to ISO 1101 / ASME Y14.5M
- Context-sensitive on-line help in all 12 User languages

Options:

- Software package CM-Surf for measuring free-form surfaces
- CAD direct interfaces (e.g. CATIA V4/V5, Pro-E, Unigraphics, Parasolid)
- I++DME Server
- DMIS Import, DMIS Export, DMIS Reporting, DMIS Native Interpreter

Machine Type		LH 1210				LH 1512			
Measuring Ranges, Dimensions, Weights									
Measuring ranges	x	[mm]	1200	1200	1200	1200	1500	1500	1500
	y	[mm]	1600	2000	2500	3000	2000	2500	4000
	z	[mm]	1000	1000	1000	1000	1200	1200	1200
Useable table surface		[mm]	1450x2715	1450x3115	1450x3615	1450x4115	1730 x 3215	1730x3715	1730x4215
Machine weight		[kg]	6000	8000	10200	12600	10800	12500	19000
Permissible part weight		[kg]	2300	2750	3000	3250	3500	3750	4250
General Requirements									
Electric		Single-phase AC 1P+N+PE, 115/230V +/- 10%, 50/60 Hz, max. 1000 VA, acc. to EN 60204/1							
Compressed air		Supply pressure 6-10 bar, pre-filtered, quality according to ISO 8573-1: Class 4 or better							
Air consumption	(Nl/min)	Min. 70, higher consumption possible, depending on application (pneum. dampers)							
Measuring Accuracy									
Measurement system		Photoelectric scale set-up system, optical division 20µm, resolution 0.1µm							
Probing uncertainty ¹	MPE _P [µm]	SP25 2.1	SP80 2.1		SP25 2.4		SP80 2.4		
Volumetric length measuring uncertainty ²	MPE _E [µm]	SP25/80 2.1+L/450				SP25/80 2.4+L/450			
Scanning probe uncertainty ³	MPE _{THP} [µm]	SP25/SP80 2.7				SP25/SP80 3.0			
Total measuring time for THP	MPT _{THP} [sec]	72				72			
Operating Environment									
Operating temperature		15°C - 30°C							
Temperature range for MPE _E		19°C - 21°C, ΔT 0.5K/h, 0.5K/m, 1K/d							
Relative humidity		40% - 70%							
Dynamics									
Joystick operation	V _{max}	0-20mm/s [creep mode], 0-100mm/s [normal]							
CNC mode	V _{max}	300mm/s axial, 520mm/s volumetric							
CNC mode	a _{max}	500mm/s axial, 800mm/s² volumetric							

1: According to DIN EN ISO 10360-2 / Maximum Permissible Error MPE_P

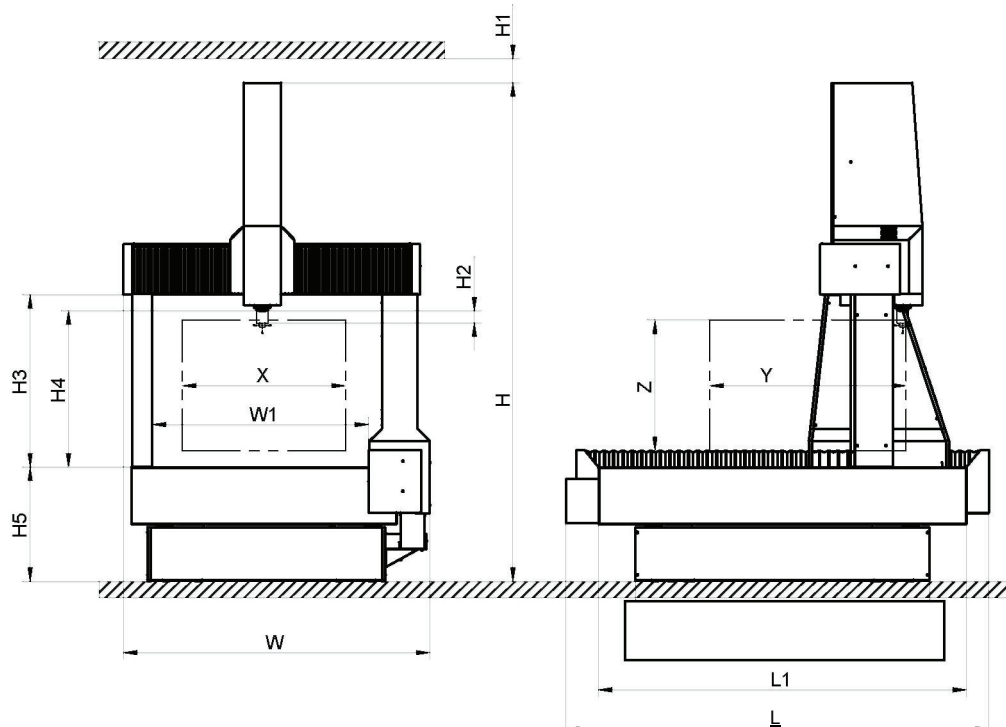
- SP25M with Module SM25-1 and Styli Ø 4 x 21 mm
- SP80 and Styli Ø 5 x 50 mm
- TP200 with Standard Force Module and Styli Ø 4 x 21 mm
- TP20 with Standard Force Module and Styli Ø 4 x 10 mm

2: According to DIN EN ISO 10360-2 / Maximum Permissible Error MPE_E

- SP25M with Module SM25-1 and Styli Ø 4 x 21 mm
- SP80 and Styli Ø 5 x 50 mm
- TP200 with Standard Force Module and Styli Ø 4 x 21 mm
- TP20 with Standard Force Module and Styli Ø 4 x 10 mm

3: According to DIN EN ISO 10360-4 / Maximum Permissible Error MPE_{THP}

- SP25M with Module SM 25-1 and Styli Ø 4 x 21 mm
- SP80 and Styli Ø 5 x 50 mm

**Overall Dimensions**

	LH 1210	LH 1512
H	3500	4100
W	2165	2580
L	3000 / 3400 / 3900 / 4400	3500 / 4000 / 4500 / 5700
X	1200	1500
Y	1600 / 2000 / 2500 / 3000	2000 / 2500 / 3000 / 4000
Z	1000	1200
H1	50	50
H2 (PH 10M)	90	90
H2 (SP 80)	145	145
H3	1165	148
H4	1150	1350
H5	700 / 700 / 660 / 710	710
W1	1465	1745
L1	2715 / 3115 / 3615 / 4115	3215 / 3715 / 4215 / 5215

All dimensions are millimetres